CDEQ V10.0: Controlling Rotating Detonation Engines Using Diesel Engine Stability Logic

Engineering Realization Based on Thermodynamic State Axioms, Countability Foundations, and Equivalent System Simplification

中文：CDEQ V10.0: 用百年柴油机的稳定逻辑控制旋转爆震发动机

—基于热力学状态公理、可数性基础与等效系统简化法的工程实现

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The DAO of Countability

Apples in a basket, one by one;
Air in a room, a mass unknown.
The countable is real, the uncountable is mist.
CDEQ builds on the real, not the mist.

Abstract

Rotating Detonation Engines (RDEs) enable high-efficiency near-constant-volume combustion. However, detonation waves rotate at microsecond periods with kilohertz pressure oscillations, creating a timescale mismatch of 3–6 orders of magnitude between the plant, sensors/actuators, and human decision cycles. By the Nyquist–Shannon theorem, direct feedback control is mathematically invalid in this regime.

A deeper issue is foundational: classical control uses continuous, uncountable mathematics, while real embedded systems are discrete and countable. This mismatch is not technical but structural.

The **CDEQ (Cross-Domain-Cyber-Dynamic-Equilibrium-Regulation-Framework)** resolves both dilemmas by migrating the century-proven stability logic of diesel engines to RDE control. Its validity rests on two axioms: the five-variable thermodynamic state set (P, T, V, S, U) and a strict countability foundation (Q14 fixed-point, finite-step verification, time layering).

The **PCTT-CW** (Penta-Cyclic-Clockwise-Topological-Theorem) proves that diesel engines and RDEs share isomorphic 5-node cyclic topologies. Using **CENHE-LEX** (Central-Holding-Nonary-Constraint-Execution-Theorem) boundary constraints and **GCVT-10** (Global-Convergence-Validation-Theorem) finite-step convergence criteria, CDEQ stabilizes RDE thrust by constraining macroscopic thermodynamic states, not tracking microsecond detonation dynamics.

A key engineering innovation is the V-shaped dual-channel 20° nozzle (20° per side, 40° total). This angle comes from the theoretical optimum of 19.1° for dual-oscillator synchronization, rounded for manufacturing. The structure achieves passive attitude stability and detonation phase locking purely through geometry.

CDEQ replaces complex algorithms with geometric constraints, microsecond tracking with state bounding, and uncountable theory with countable implementation. It follows a Result-to-Cause design philosophy: define the stable output first, then derive the necessary constraints.

Keywords:

Rotating Detonation Engine; RDE; CDEQ; timescale mismatch; countability; five-variable thermodynamics; V-shaped dual-channel nozzle; 20° nozzle; self-stabilizing structure

Fundamental Premises

CDEQ is built on three irrefutable physical facts:

1. RDE state evolution is intrinsically microsecond-scale, fixed by thermochemistry.
2. Human and digital controller cycles have hard upper limits.
3. Direct feedback across 3+ orders of magnitude timescale mismatch is information-theoretically impossible.

Thus, any scheme chasing microsecond detonation dynamics is infeasible. CDEQ is the only viable architecture under these constraints.

1. Problem: The Dual Dilemma of Cybernetics

1.1 RDE Physics and Status Quo

RDEs use self-sustaining annular detonation waves with 1500–2000 m/s rotation, 15–30 MPa pressure, and kHz frequencies. These dynamics are physically fixed at the microsecond scale.

Global research has achieved progress but not fundamental stability. All existing methods face the same timescale barrier.

1.2 Core Contradiction: Timescale Mismatch

The Nyquist–Shannon theorem requires the control cycle to be faster than half the plant characteristic period. For RDEs, this is impossible with existing hardware:

- Detonation: microseconds
- Sensors/actuators: milliseconds
- Human decisions: 167 ms

Control commands always lag state evolution, guaranteeing instability or overload.

1.3 Foundational Crisis: Uncountable vs. Countable

Classical control uses continuous, uncountable mathematics (real numbers, differentials, limits, infinite convergence).
Real systems are discrete and countable: finite states, fixed-point arithmetic, clock cycles, bounded memory.

This mismatch makes formal verification and deterministic implementation impossible.

1.4 Limits of Traditional Approaches

Active control methods (plasma, fuel modulation) cannot overcome timescale limits. Passive methods (combined chambers) reduce fluctuations but do not guarantee stability. All focus on detonation dynamics, not stable thrust.

1.5 Conclusion: A New Paradigm

We must stabilize output states, not track internal processes.
We must rebuild control on countable mathematics, not continuous analysis.
This is the CDEQ paradigm.

This work is built upon the six foundational charters of the CDEQ framework, which have been legally registered with trusted TSA timestamps (see References [20]–[25]). The complete evidence chain predates this paper and establishes sole authorship and originality.

1.6 A Simple Metaphor: The Cat That Comes Back

Traditional control theory attempts to track microsecond detonation waves with millisecond response cycles—like chasing a cat that runs faster the harder you pursue.

The CDEQ framework takes the opposite approach: stop chasing the cat. Instead, draw a rigid boundary, sit calmly with tea and newspaper, and wait for the cat to return on its own.

This is not an optimization of existing control strategies, but a fundamental shift in control philosophy: from chasing dynamics to locking boundaries, from real-time feedback to pre-defined containment.

The cat will always come back—because the boundary leaves it nowhere else to go.

2. Countability Foundation

2.1 Countable vs. Uncountable

A set is countable if its elements can be mapped one-to-one with natural numbers.

- Countable: integers, fixed-point values, finite states
- Uncountable: real numbers, continuous trajectories

2.2 Intuitive Metaphor

Countable = apples in a basket (definite, verifiable).
Uncountable = air in a room (ambiguous, immeasurable).
CDEQ works only with countable quantities.

2.3 Q14 Fixed-Point Standard

CDEQ uses a single unified fixed-point format named Q14:

- Q14_1P0 = 16384: Q14 fixed-point reference representing 1.0
- clip: clamping function that restricts values to [0, 16383]
- Range: $0 \leq \text{value} \leq 16383$
- Exactly 16384 countable states

All variables use Q14, eliminating ambiguity, conversion error, and nondeterminism.

2.4 Philosophy of One Format

One format = one rule set = perfect alignment between theory and code.
Continuous physical quantities become discrete, countable, verifiable values.

3. Methodology: Equivalent System Simplification

CDEQ uses equivalent system simplification: map the complex RDE to a well-understood, stable system (the diesel engine) and migrate its control logic.

This is analogous to Cao Chong weighing an elephant: replace the elephant with equivalent stones, then weigh the stones.

Diesel engines provide three critical properties: determinism, robustness, century-proven stability. Their control logic is migrated to the RDE via topological isomorphism.

CDEQ follows Result-to-Cause design:
Define stable thrust first → derive constraints → enforce them.

4. Principle: Dynamical System Isomorphism

Diesel engines and RDEs appear drastically different but share the same cyclic 5-node thermodynamic topology, proven by the PCTT-CW (Penta-Cyclic-Clockwise-Topological-Theorem).

This is like a bicycle and a motorcycle: different speeds, same balancing principle.

PCTT-CW shows that stable cyclic systems require a 5-node closed loop:

- 3 nodes: divergence
- 4 nodes: deadlock
- 5 nodes: balanced generation and suppression
- ≥6 nodes: chaos and inefficiency

Thus, diesel engine control strategies can be isomorphically migrated to RDEs, with only timescale scaling.

Lyapunov's theory tells us what stability means. CDEQ tells us how to build it.

5. Core State Variables: Five-Variable Thermodynamic Closed Space

All engine states are uniquely defined by five macroscopic variables:

- Pressure P
- Temperature T
- Volume V
- Entropy S
- Internal Energy U

These form a closed, axiomatic thermodynamic system, inherited from classical thermodynamics (Carnot, Clausius, Kelvin, Gibbs, Wang Zhuxi).

CDEQ constrains these macroscopic averages, not microscopic detonation dynamics.

All values use Q14 fixed-point for countability and determinism.

6. The Three Core Theorems

6.1 PCTT-CW: Penta-Cyclic-Clockwise-Topological-Theorem

Nodes: P_0, P_1, P_2, P_3, P_4 (Q14 state variables)

- φ: forward excitation factor = 0.618
- $\bar{\varphi}$: reverse damping factor = 0.382
- S = 10124: Q14 engineering value of φ
- C = 6258: Q14 engineering value of $\bar{\varphi}$
- Ψ': system state anchoring variable
- Ψ_{-1}: previous state
- Ψ_{-2}: state two steps before
- ω: chaotic weight coefficient
- ξ: anti-deadlock perturbation

V_PCTT (PCTT-CW deviation function):
$V_PCTT = \Sigma_i \left(P_i^2 - (S/16384)\cdot P_i \cdot P_{i-1} + (C/16384)\cdot P_i \cdot P_{i-2} \right)$

Update rule:
$P_i = \text{clip}\left(P_i + (S\cdot P_{i-1} - C\cdot P_{i-2}) / 16384 , 0, 16383 \right)$

This is the only integer pair in Q14 for the golden ratio.
The system forms a bounded, convergent 5-node cycle.

6.2 CENHE-LEX: Central-Holding-Nonary-Constraint-Execution-Theorem

3×3 nine-field matrix E with Q14 elements.

- $E_{11} = 5461$: fixed central anchor value
- R_i: row sum
- C_j: column sum
- D_0: main diagonal sum
- D_1: anti-diagonal sum
- ∇C: deviation correction operator

V_CENHE (CENHE-LEX deviation function):
$V_CENHE = [\Sigma_i(R_i-16384)^2 + \Sigma_j(C_j-16384)^2 + (D_0-16384)^2 + (D_1-16384)^2] / 16384$

This enforces energy conservation and central stability.

6.3 GCVT-10: Global-Convergence-Validation-Theorem

Total deviation function:
$V(t) = V_PCTT(t) + V_CENHE(t)$

- $\varepsilon = 1636$: E-version error threshold (10% of Q14 full scale)

- ∀10: 10 consecutive steps validation rule
- TAI: stable steady state after convergence

Stability criterion (deviation, V function):
$V(t) < \varepsilon$ for 10 consecutive steps.
$V \geq 0$ at all times.
V keeps decreasing toward convergence.
Stability is achieved when deviation reaches zero.

GCVT-10 is an engineering acceptance rule, not an abstract mathematical definition. It answers:
“Is the system safe now, within finite time?”

Lyapunov's theory tells us what stability means. CDEQ tells us how to build it.

6.4 Synergy

- PCTT-CW: dynamics
- CENHE-LEX: boundaries
- GCVT-10: verification

Together they form the CDEQ core.

7. Time-Layered Architecture

CDEQ defines six Timescale Layers (TLS):

- TLS0: 167 ms (observer)
- TLS1: ms (industry)
- TLS2: 100 μs (flight)
- TLS3: 10 μs (hypersonic)
- TLS4: μs (detonation)
- TLS5: ns/ps (nuclear)

RDEs run at TLS4; observers at TLS0.
The principle:
Use light to control light, use fast to control fast, use slow to govern fast.

- TLS4: hardware closed-loop (FPGA, sensors)
- TLS0: human acceptance (167 ms reports)

No direct cross-layer control. Layers operate autonomously.

8. Core Structure: V-Shaped Dual-Channel 20° Nozzle

8.1 Geometry

- Two symmetric channels, 20° from axis each
- Total included angle: 40°

8.2 Origin

Theoretical optimum: $\theta_0 = 19.1°$ (from dual-oscillator synchronization).
Rounded to 20° for engineering.
Performance loss < 1.5%.

8.3 Mechanism

1. Lateral force cancellation: left/right lateral components cancel internally.
2. Phase locking: 40° geometry synchronizes detonation waves passively.

No sensors, no algorithms, no flight control needed. Stability is structural.

8.4 Tradeoff

~6% thrust loss for:

- Zero lateral load
- Zero yaw correction
- Intrinsic stability
- Extreme robustness

This is a system-level net gain.

9. Core Master Equation

$$\Psi' = \mathrm{clip}(\, \varphi \cdot \Psi_{-1} - \bar{\varphi} \cdot \Psi_{-2} + \omega \cdot \xi \,) + (\, \Psi \geq \Psi\pi \,) \cdot (\, \nabla C + \kappa \cdot \Delta x \cdot \cos 20° \,)$$

Unifies:

- PCTT-CW evolution
- CENHE-LEX central return
- 20° geometric coupling
- Q14 clipping
- Finite-step verification

10. Stance: Thrust Is Fundamental

Advanced tools are worthless if unstable.
Simple methods are truth if they work.

Thrust is fundamental; process is secondary.
Result is heavy; showmanship is light.

CDEQ is engineering, not spectacle: usable, reliable, solid.

11. Summary

CDEQ V10.0 migrates diesel engine stability logic to RDEs using:

1. Five-variable thermodynamics
2. PCTT-CW / CENHE-LEX / GCVT-10
3. Q14 countable fixed-point
4. Time-layered autonomy
5. 20° V-shaped self-stabilizing nozzle

It resolves the timescale mismatch and foundational mathematical crisis simultaneously.

Fire in the hole.

Appendix A: Angle Verification

Only 20° balances thrust, coupling, and GCVT-10 convergence.
0–10°: too weak
30°+: excessive loss
40°+: unstable

Appendix B: Classical Origins

CDEQ is built on:

- Thermodynamics (Carnot → Wang Zhuxi)
- Information theory (Nyquist–Shannon)
- Cybernetics (equivalent systems)
- Graph theory (C_5 cycle)
- Stability theory (Lyapunov)
- Countability foundations

Appendix C: Q14 Specification

- Name: CDEQ-Q14
- Type: uint16_t
- Q14_1P0 = 16384
- Range: 0–16383
- clip: [0, 16383]
- Golden coefficients: S=10124, C=6258 (unique pair)

Intellectual Property Statement

All theoretical frameworks, formulas, algorithms, codes, and structural designs in this paper are the sole and original intellectual property of Jiang Wenjia. The complete manuscript has been legally registered and certified with a trusted TSA timestamp prior to submission. Any unauthorized use, plagiarism, misappropriation, or authorship falsification will result in immediate and full legal action.

This paper is the foundational charter of the CDEQ framework. All subsequent versions and derivative works must cite this original source.

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The six foundational charters serve as the unique verification code of originality. They are the cat at the door — only the true owner knows its name. No counterfeit can pass the check.

This is not a warning. This is a blade.
If you reach for the fruit, expect to lose the hand.

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